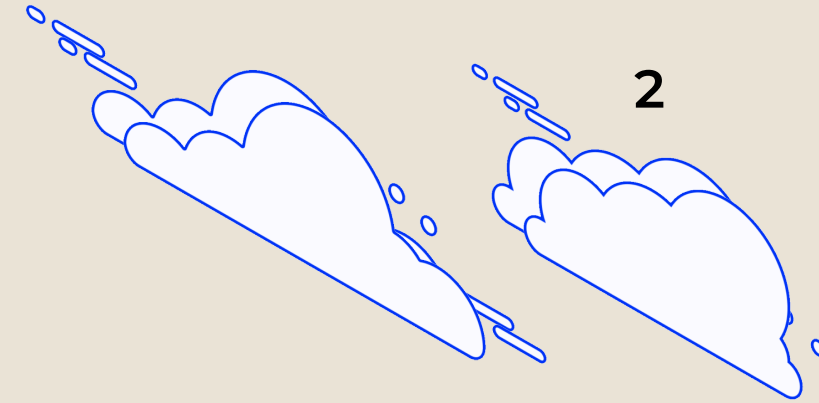


SENTIMENT ANALYSIS FOR REAL-TIME CUSTOMER FEEDBACK IN E-COMMERCE



By : Nisarga Shivamurthy Maheshwarappa

TABLE OF CONTENTS



2

- **Introduction**

- **Research Methodology**

- **Summary and findings**

- **Problems faced**

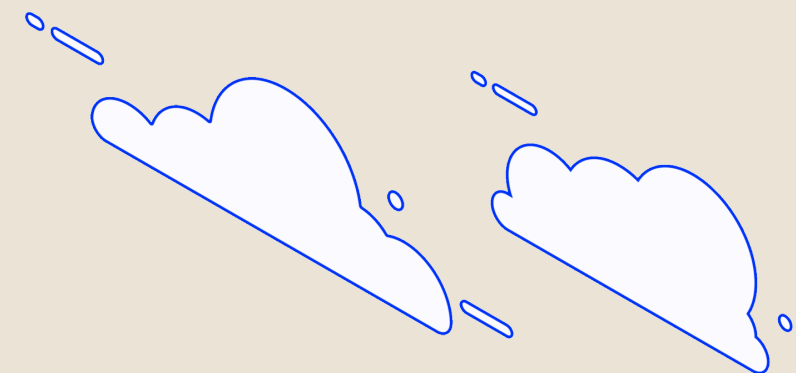
- **Machine Learning Models**

- **Chatbot**

- **Hybrid Sentiment Analysis Pipeline Using OpenAI & ChromaDB**

- **Conclusion**

- **References**



Introduction



- E-commerce has transformed how businesses interact with customers in the digital era.
- Online reviews play a crucial role in influencing potential buyers and helping businesses improve products/services.
- Customers frequently express their opinions via reviews, ratings, and comments.
- These opinions are rich sources of unstructured data that can reveal valuable consumer insights.
- Sentiment Analysis, a branch of Natural Language Processing (NLP), helps classify these opinions into positive, negative, or neutral sentiments.

Aim and Objectives



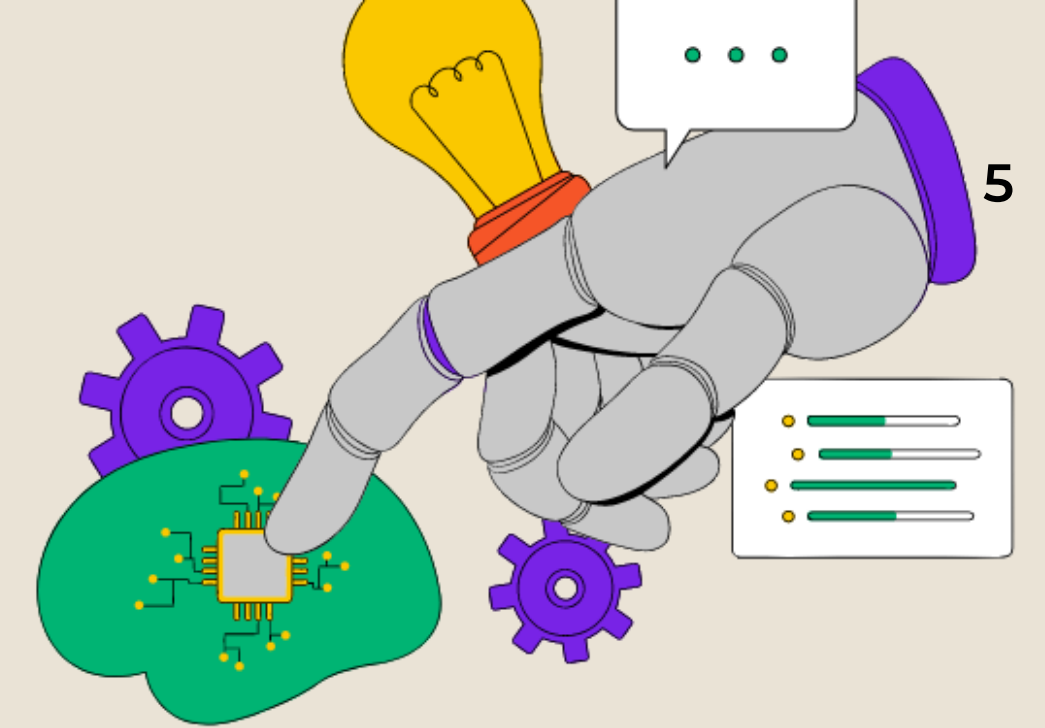
Aim

To implement a sentiment analysis model that automatically classifies e-commerce customer feedback as positive, negative, or neutral, enabling real-time insights and proactive decision-making to enhance service quality and customer satisfaction.

Objectives

- Develop a robust sentiment analysis model using state-of-the-art NLP techniques.
- Evaluate and compare traditional ML models (Naïve Bayes, Logistic Regression) with deep learning models (BERT, RNN-LSTM) using metrics like accuracy, precision, recall, and F1 score.
- Analyze sentiment trends across major product categories (e.g., laptops, mobile phones, TVs) to uncover emotional triggers in customer feedback.
- Extract actionable insights from sentiment outcomes to support product recommendations, improve customer engagement, and inform marketing strategies.

Problem Statement



RQ1: How to effectively apply sentiment analysis to customer reviews in e-commerce?

RQ2: Which machine learning and deep learning models give the maximum possible accuracy for classifying customer sentiments?

RQ3: How does sentiment analysis improve the functionality of systems such as recommendation engines or chatbots in e-commerce?

RQ4: What are the possible issues that could impede the use of sentiment analysis in the e-commerce review context? How can they be solved?

Methodology

6

1. Business Understanding

- Identified the real-world problem: improving customer satisfaction by analyzing e-commerce feedback in real time.
- Defined project goals and success metrics clearly.

2. Data Understanding

- Explored Amazon customer review data (text + star ratings).
- Analyzed patterns, trends, and data quality issues in customer feedback.

3. Data Preparation

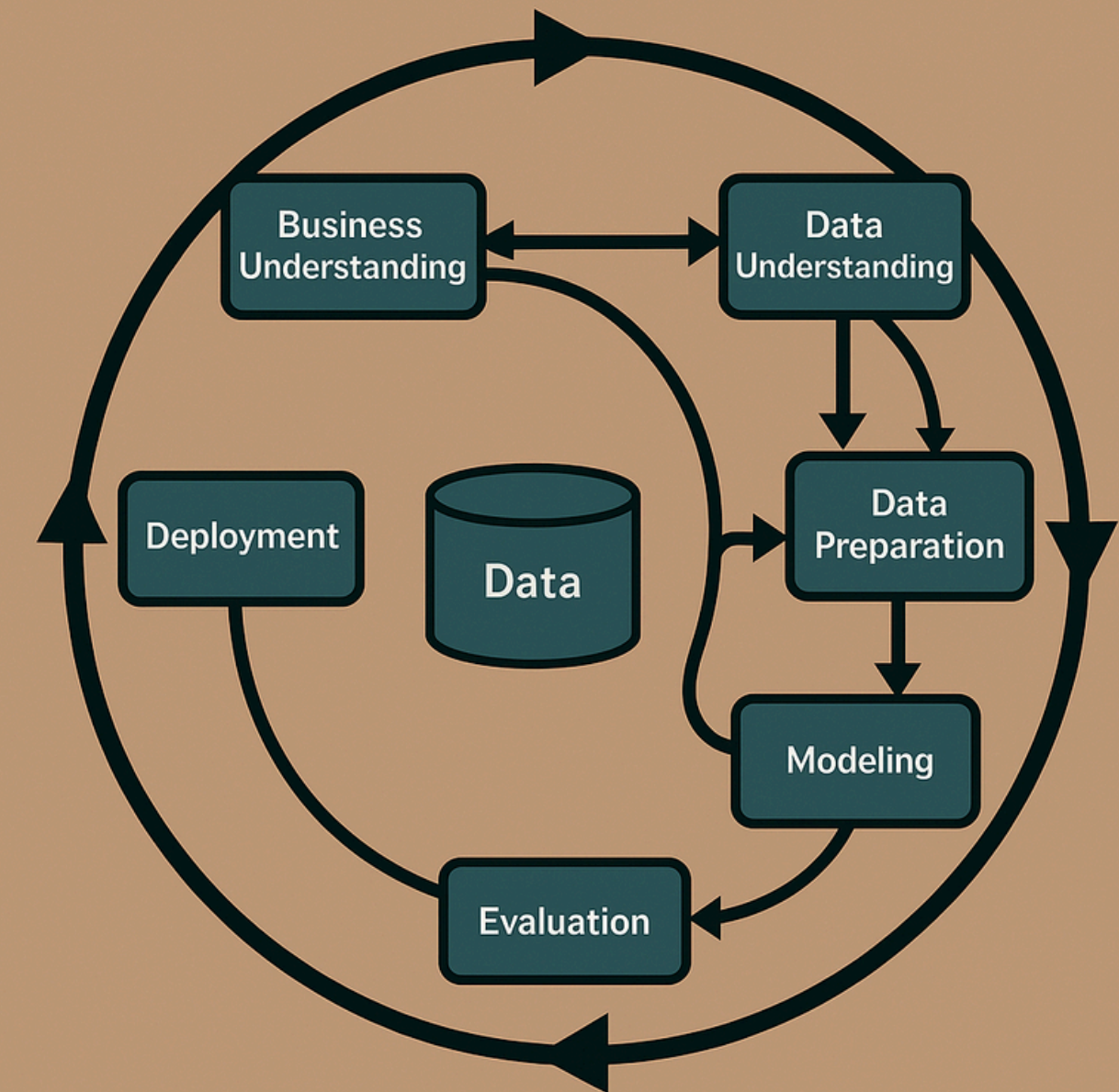
- Cleaned and preprocessed text data (tokenization, stopwords removal, normalization).
- Labeled data for sentiment classification (Positive, Negative, Neutral).

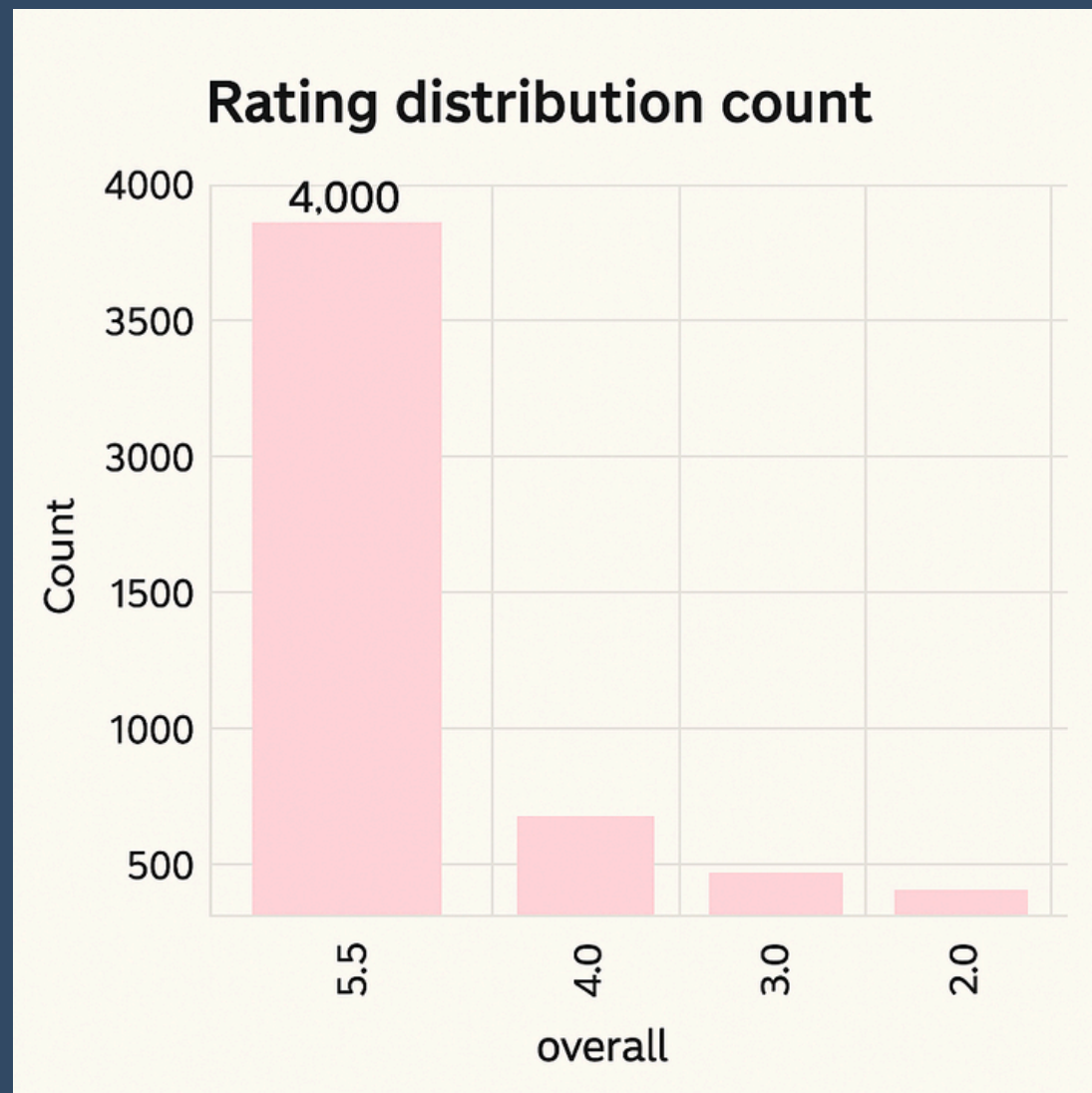
4. Modeling

- Trained various models: Logistic Regression, Naïve Bayes, LSTM.
- Used NLP techniques to classify sentiments based on textual data.

5. Evaluation

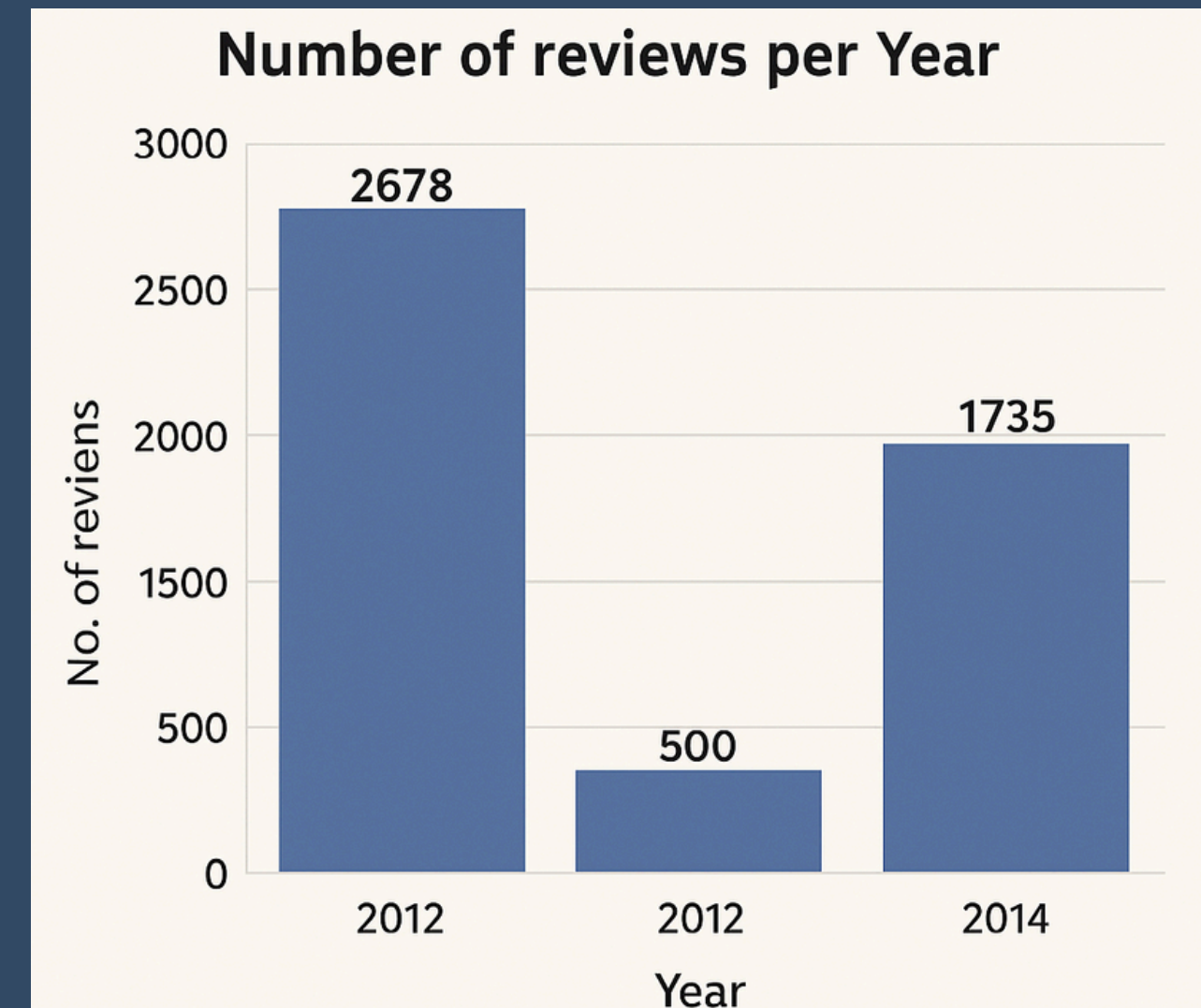
- Compared models using accuracy, precision, recall, and F1 score.
- Ensured model performance aligns with the business goal of reliable sentiment detection.





Count plot for Overall

The bar chart visualizes the distribution of product ratings in the dataset. It clearly shows that most customers gave a 5-star rating, indicating a strong positive bias in the feedback. Lower ratings like 1, 2, and 3 stars occur much less frequently, highlighting an imbalanced rating pattern.



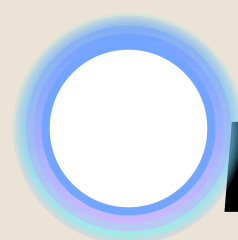
Bar plot for Year and Number of Reviews

This bar chart shows the number of customer reviews per year, highlighting temporal trends in review activity. The year 2013 recorded the highest number of reviews (2678), followed by 2014 with 1735, while 2012 saw significantly fewer reviews (500). This indicates that customer engagement peaked in 2013, providing a dense period of data for sentiment analysis.

Problems Faced

9

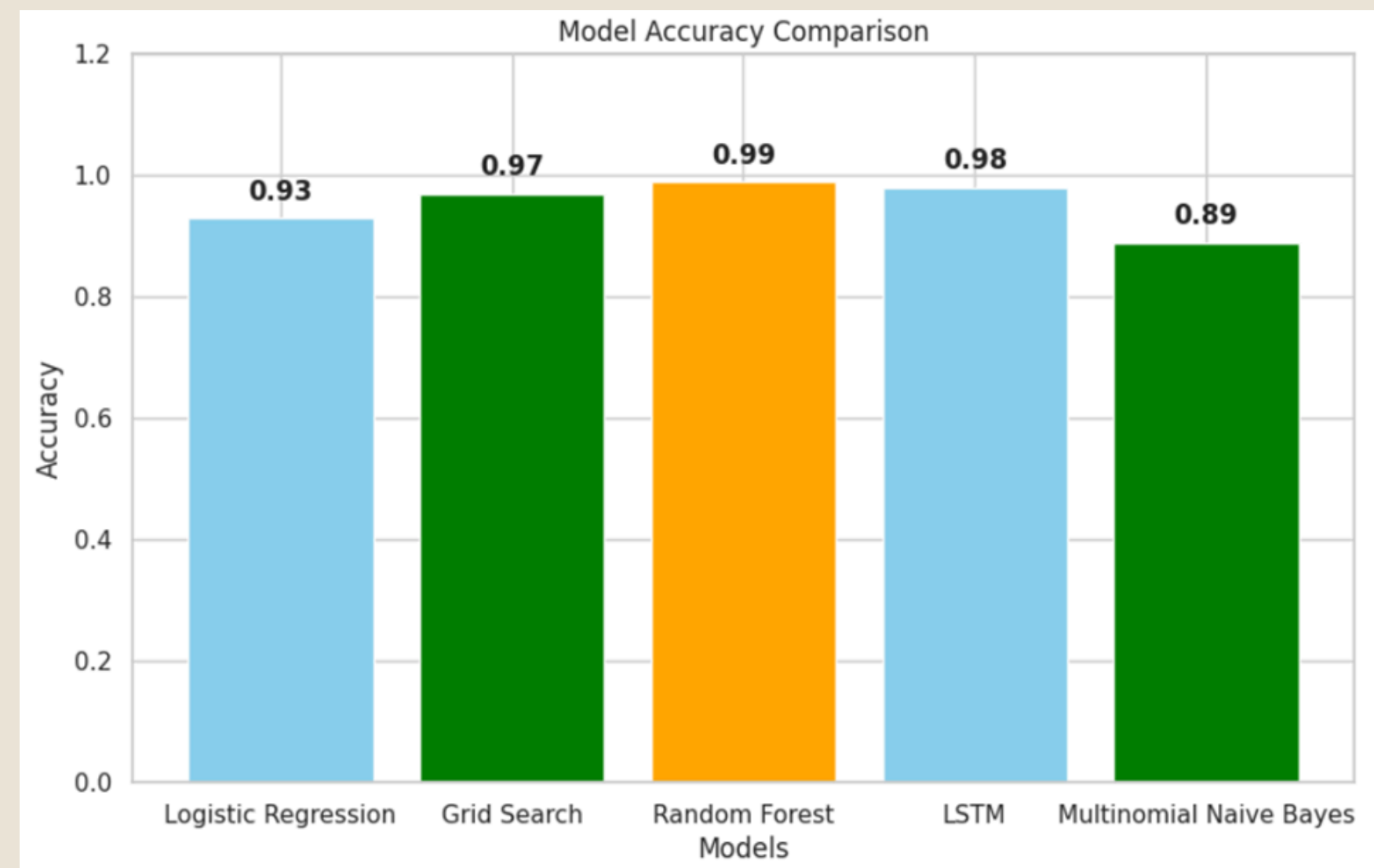
- All the data in reviews will not be balanced : implimented sampling technique to overcome and balance the data
- Automatically generated or biased reviews can skew the sentiment distribution and confuse the model.
- Longer reviews may contain multiple sentiments or topics, making a single-label classification insufficient.
- Misspellings, emojis, slang, special characters, or inconsistent punctuation
- Ambiguity and Mixed Sentiment Example: "The sound quality is amazing, but the battery life is terrible."
- Sarcasm and Irony Example: "Oh great, another product that broke in a day. Just what I needed!"
- 3 stars could be neutral or slightly positive depending on the user :
usedvader_lexicon to label the reviews rather than depending on the ratings



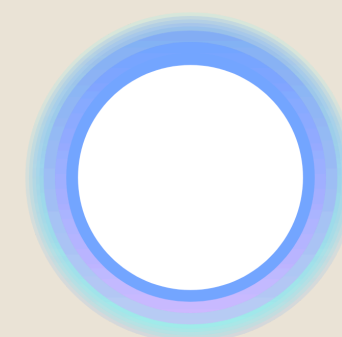
Machine Learning Models for Sentiment Classification

ML Pipeline Overview:

1. Data Cleaning: Removed noise and irrelevant tokens
2. Data Imbalance: Detected imbalance → Applied data balancing
3. Artificial Review Generation: Used OpenAI to generate synthetic reviews
4. Logistic Regression: Baseline model
 - Improved with GridSearchCV for hyperparameter tuning
5. Random Forest: Achieved highest accuracy
6. Multinomial Naive Bayes: Tried for comparison
7. LSTM Model (RNN): Implemented using Keras Sequential model

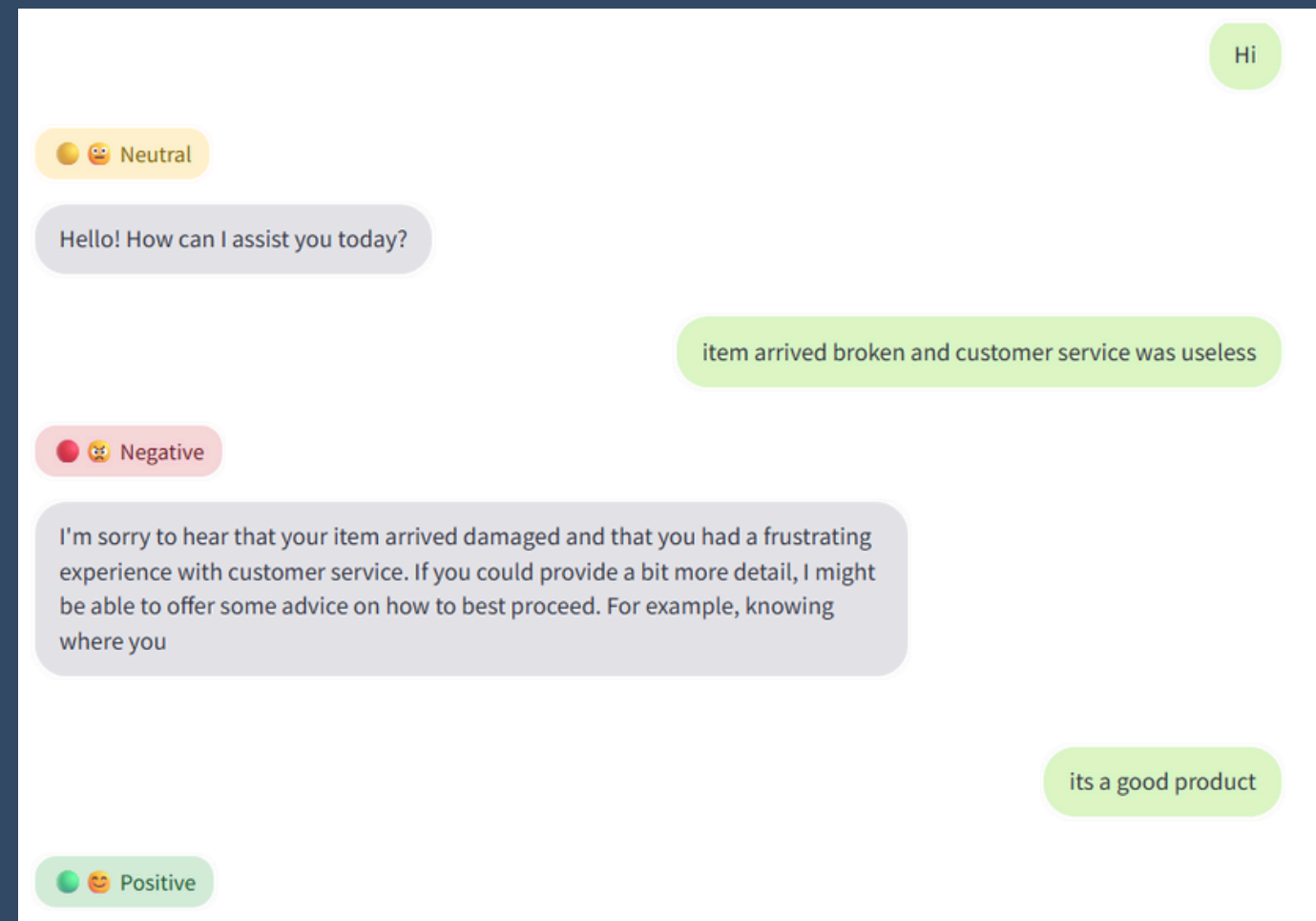


Models

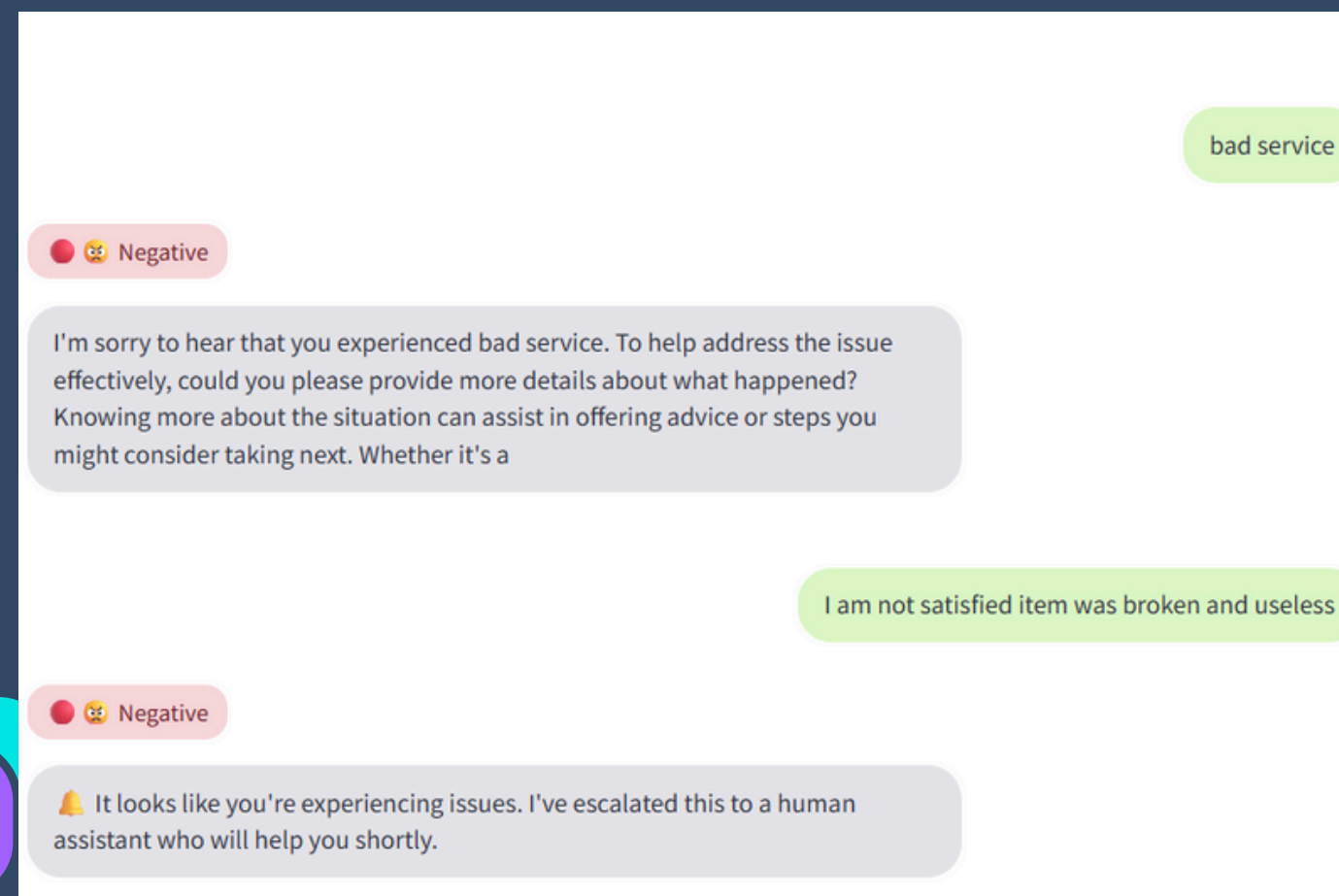


Chat Bot

11



- This is an AI-powered customer service chat interface that analyzes user sentiment in real time using a machine learning model.
- Messages are visually styled like WhatsApp, and each user input is tagged with a sentiment (Positive, Neutral, Negative).



- User sends message
- Bot processes message and performs sentiment analysis
- If negative sentiments occur consecutively, the system automatically escalates the issue to a human assistant for further support.
- Else, bot continues the conversation



Hybrid Sentiment Analysis Pipeline

Using OpenAI & ChromaDB



Workflow

1. CSV Dataset → Pandas
2. OpenAI Embedding (text-embedding-ada-002)
3. ChromaDB (Vector DB)
4. Similarity Search (Top 5 reviews)
5. GPT-4 Prompt → Predicted Sentiment



What We Did:

- Converted reviews into dense embeddings
- Stored in ChromaDB with sentiment labels
- Retrieved most similar reviews to a new query
- Prompted GPT-4 with similar examples for sentiment inference

Why This Works:

- Combines semantic understanding with generative reasoning
- Uses real examples for better contextual sentiment analysis
- Flexible, explainable, and scalable

Conclusion

- Successfully implemented sentiment analysis on Amazon reviews using both ML (Logistic Regression, Naive Bayes, Random Forest) and DL (LSTM) models.
- CRISP-DM methodology guided systematic data sampling, cleaning, balancing, and transformation (TF-IDF, GPT-generated neutral reviews).
- Random Forest achieved the highest accuracy (0.99) but showed signs of overfitting and poor generalization.
- Logistic Regression with Grid Search offered more interpretable and balanced performance (0.97 accuracy), making it suitable for real-world use.
- Naive Bayes underperformed due to strong independence assumptions that don't align with natural language context.
- Developed two Streamlit chatbot applications for real-time sentiment detection using:
 - Logistic Regression (for fast predictions)
 - RAG-based bot (with OpenAI & ChromaDB embeddings for richer responses)
- Challenges like class imbalance, sarcasm, and mixed sentiment were mitigated via strong preprocessing and synthetic data generation.
- Highlights the practical impact of sentiment analysis in enhancing product recommendations, human-like chatbot interactions, and customer service.
- Concludes that simple, well-tuned models can outperform complex ones in real-time applications and improve customer engagement effectively.



Future Scope

- Move beyond basic sentiment (positive/neutral/negative) to detect emotional tones like anger, joy, disappointment, etc.
- Leverage advanced NLP models like BERT and RoBERTa for better context understanding and multilingual capabilities.
- Integrate sentiment models into chatbots and support tools to deliver more empathetic and adaptive customer interactions.
- Explore Cross-Domain Learning and Transfer Learning to enhance model generalization across diverse data sources.
- Adopt Explainable AI (XAI) to improve transparency and user trust in model predictions.
- Utilize additional data sources (e.g., social media, service logs, multilingual reviews) for deeper emotional insights.
- Develop mobile-friendly and offline-capable chatbot versions for wider accessibility.
- Combine sentiment analysis with collaborative filtering to build hybrid recommendation systems for hyper-personalized shopping experiences.

References

- Muhammad Haroon¹, Zaheer Alam², Rukhsana Kousar³, Jawad Ahmad⁴, Fawad Nasim” Sentiment Analysis of Customer Reviews on E-commerce Platforms: A Machine Learning Approach” Bulletin of Business and Economics, 13(3), 230-238
- Davoodi, L., Mezei, J. & Heikkilä, M. Aspect-based sentiment classification of user reviews to understand customer satisfaction of e-commerce platforms. Electron Commer Res (2025). <https://doi.org/10.1007/s10660-025-09948-4>
- Jahanzeb Jabbar, Iqra Urooj , Wu JunSheng. Naqash Azeem, “Real-time Sentiment Analysis On E-Commerce Application”, Proceedings of the 2019 IEEE 16th International Conference on Networking, Sensing and Control, May 9-11, 2019, Banff, Alberta, Canada
- Bellar, Oumaima, Amine Baina, and Mostafa Ballafkih. 2024. "Sentiment Analysis: Predicting Product Reviews for E-Commerce Recommendations Using Deep Learning and Transformers" Mathematics 12, no. 15: 2403. <https://doi.org/10.3390/math12152403>
- Huang, H., Zavareh, A.A. and Mustafa, M.B., 2023. Sentiment analysis in e-commerce platforms: A review of current techniques and future directions. IEEE Access, 11, pp.90367-90382.
- Alzahrani, M.E., Aldhyani, T.H., Alsubari, S.N., Althobaiti, M.M. and Fahad, A., 2022. Developing an Intelligent System with Deep Learning Algorithms for Sentiment Analysis of E-Commerce Product Reviews. Computational Intelligence and Neuroscience, 2022(1), p.3840071.
- Patra, G.K., Kuraku, C., Konkimalla, S., Boddapati, V.N. and Sarisa, M., 2023. Sentiment Analysis of Customer Product Review Based on Machine Learning Techniques in E-Commerce. Journal of Artificial Intelligence & Cloud Computing. SRC/JAICC-408. DOI: [doi. org/10.47363/JAICC/2023](https://doi.org/10.47363/JAICC/2023) (2), 389(1), pp.7211-7224.

Thank
you!

